HANZHI YANG

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EDUCATION

Doctor of Philosophy August 2022 - Present
Purdue University, GPA: 3.9

Master of Science in Engineering August 2020 - May 2022
University of Michigan, GPA: 3.92

Bachelor of Science in Mechanical Engineering August 2016 - May 2020
Purdue University, GPA: 3.9

RESEARCH INTERESTS

Robotics, Autonomous Systems, Nonlinear Dynamics, Robust Control, Intelligent Control

EXPERIENCE

Research Assistant August 2022 - Present
Mahmoudian Lab, Purdue University

Teaching Assistant August 2025 - December 2025
Autonomous System, Purdue University

Research Associate January 2021 - February 2022
Kurabayashi Group, University of Michigan

Research Assistant May 2021 - September 2021
Transportation Research Institude, University of Michigan

AWARDS AND HONORS

John P. Sharp Memorial Fellowship 2022

William H. and E. Jean Pfaff Scholarship in Mechanical Engineering 2018 & 2019

RESEARCH PUBLICATIONS

Yang, H., & Mahmoudian, N. (2025). Finite-time prescribed performance with fixed-time disturbance rejection for underwater glider heading control. Ocean Engineering, 337, 121842. October 2025

Yang, H., & Mahmoudian, N. (2024). Under-ice trajectory tracking for underwater gliders using fuzzy-based adaptive control. In 2024 IEEE/OES Autonomous Underwater Vehicles Symposium September 2024

Yang, H., & Mahmoudian, N. (2024). Gliding in extreme waters: Dynamic Modeling and Nonlinear Control of an Agile Underwater Glider. IFAC-PapersOnLine, 58(20), 121-126. September 2024

Yang, H., Yu, J., Birla, M. B., Wang, T. D., & Oldham, K. R. (2022). Linear Modeling and Open-loop Control of a Multi-Axis Piezoelectric Micro-Mirror for Random-Access Laser Scanning. IFAC-PapersOnLine, 55(37), 119-124. October 2022

Kusari, A., Li, P., **Yang, H.**, Punshi, N., Rasulis, M., Bogard, S., & LeBlanc, D. J. (2022). Enhancing SUMO simulator for simulation based testing and validation of autonomous vehicles. In 2022 ieee intelligent vehicles symposium (IV) (pp. 829-835). IEEE. June 2022